

Final

Spring - 2018

①

①

②

```
#include <stdio.h>
```

```
void main() {
```

```
int n;
```

```
scanf ("%d", &n);
```

```
for (int i = 1; i ≤ n; i++) {
```

```
    for (int j = 1; j ≤ n - i + 1; j++) {
```

```
        printf ("*");
```

```
    }
```

```
    for (int k = 1; k ≤ i - 1; k++) {
```

```
        printf (" ");
```

```
    }
```

```
    printf ("\n");
```

```
}
```

```
}
```

②

⑥

step 1:

$n = 3$

$i = 0; i < 3$

$j = 0; j \leq 0$

$sum = 0 + 0 + 0 = 0$

step 2

$i = 1; i < 3$

$j = 0; j \leq 1$

$sum += i + j$

$= 0 + 1 + 3 = 4$

step 3

$i = 1; i < 3$

$j = 1; j \leq 1$

$sum += i + j$

$= 4 + 1 + 1$

$= 6$

step 4

$i = 2; i < 3$

$j = 0; j \leq 2$

$sum += i + j$

$= 6 + 2 + 3 = 11$

step 5

$i = 2; i < 3$

$j = 1; j \leq 2$

$sum += i + j$

$= 11 + 2 + 3 = 16$

step 6

$i = 2; i < 3$

$j = 2; j \leq 2$

$sum += i + j$

$= 16 + 2 + 2 = 20$

②

#include <stdio.h>

typedef struct {

char name[50];

char phnNum[20];

char address[30];

} phone book;

void main {

phonebook p[100]

<1.0162> shubam #

for (int i=0; i<100; i++) {

scanf ("%s", p[i].name);

scanf ("%s", p[i].phnNum);

} scanf ("%s", p[i].address)

char searchNam[50];

scanf ("%s", searchNam);

for (int i=0; i<100; i++) {

if (strcmp (p[i].name, searchNam) == 0) {

printf ("%s %s", p[i].address, p[i].phnNum);

break;

}

}

if (i == 100) {

printf ("No data found");

④

⑩③

①

#include <stdio.h>

void main() {

int m, n;

scanf("%d %d", &m, &n);

int arr[m][n];

for (int i = 0; i < m; i++) {

for (int j = 0; j < n; j++) {

scanf("%d", &arr[i][j]);

}

for (int i = m - 1; i >= 0; i--) {

for (int j = n - 1; j >= 0; j--) {

printf("%d", arr[i][j]);

}

printf("\n");

}

Take
input

⑥

4 7 3 7 (n) digit (n)
 5 3 8 2
 2 6 2 4 10 = sum (n)
 6 4 4 6 (0 ≠ n) while

⑤

04

②

#include <stdio.h>

int sumDigit (int);

void main () {

int n;

scanf ("%d", &n);

int ans = sumDigit (n);

printf ("%d", ans);

}

(int sumDigit (int n) {

int sum = 0;

while (n ≠ 0) {

sum += n

⑥

```
int sumDigit (int n) {
```

```
    int sum = 0;
```

```
    while (n != 0) {
```

```
        int rem = n % 10;
```

```
        sum += rem;
```

```
        n = n / 10;
```

```
    }
```

```
    return sum;
```

```
}
```

```
}
```

⑥

$x = \text{func1}(5)$

$= 10 = 5 + 5$

$y = \text{func2}(3)$

$= 10 + 3 + 8$

$= 21$

$a = 5$
 $i = 5$

$a = 10$
 $i = 3$

$x = 8$

$\text{func1}(3)$

$a = 5$
 $i = 3$

5

a

$p > n$

$a = 1 \rightarrow \text{strcmp}(str1, str2)$

$b = -1 \rightarrow \text{strcmp}(str2, str1)$

$n < p$

$str1 \rightarrow$ String Copy (Ans)

b

#include <stdio.h>

void main()

char str1[100], str2[100];

scanf("%s %s", str1, str2);

int i;

for (i = 0; str1[i] != '\0'; i++)

for (int j = 0; str2[j] != '\0'; j++)

str1[i+j] = str2[j];

}

printf("%s", str1);

}

⑧

Fall-19

①

①

step 1

$a = 11, b = 25$

print $\rightarrow b, a = 25, 11$

step 2

goto

func1 (5.5)

print $\rightarrow 5.500000$

step 3

call func2 (12, 15.0)

print $\rightarrow x, y = 12, 15.000000$

step 5

back to main

$a = \text{func1}(5.5)$

$b = 25 * 11$

$= 275$

return 274

step - 4

call func2(5, 4.5)

print $\rightarrow 5, 4.500000$

Print $\rightarrow a, b = 274, 275$

⑥

#include <stdio.h>

int digitSum(int);

int digitPos(int);

Void main()

int n;

scanf("%d", &n);

int sum = digitSum(n);

int pos = digitPos(n);

printf ("%d %d", sum, pos);

}

int digitSum (int n) {

int sum = 0;

while (n != 0) {

int rem = n % 10;

sum += rem;

n = n / 10;

}

return sum;

}

int digitPos (int n) {

int count = 0;

while (n != 0) {

count ++;

n = n / 10;

return count;

(e) sum

e ← temp

(e) sum

e ← temp

(i) sum

e = x

e ← temp

8. वृत्तान्ति ना जी

digit Count वाक दिखस

10

2

Q

func(3)

print → 3

func(2)

print → 2

func(i) → x

x = 1

↳ flow of program return from
where function was called (LIFO)

print → 2
3

~ 3 2 2 3

6

```
#include <stdio.h>
```

```
int arrSum(int[], int);
```

```
void main() {
```

```
    int n, arr[100];
```

```
    scanf("%d", &n);
```



```
for (int i=0; i < n; i++) {  
    scanf ("%d", &arr[i]);  
}
```

```
int sum = arrSum (arr, n);  
printf ("%d", sum);  
}
```

```
int arrSum (int arr[], int n) {
```

```
    if (n == 0) {
```

```
        return;
```

```
    }
```

```
    else {
```

```
        return arr[n-1] + arrSum (arr, n-1);
```

```
    }
```

```
}
```



12

3

```
#include <stdio.h>
```

```
typedef struct {
```

```
    char name[20];
```

```
    int id;
```

```
    float marks;
```

```
} student;
```

```
void main() {
```

```
    student s;
```

```
    for (int i=0; i<4; i++) {
```

```
        scanf ("%s", s[i].name);
```

```
        scanf ("%d", &s[i].id);
```

```
        scanf ("%f", &s[i].marks);
```

```
    float minMark = s[0].marks;
```

```
    int minIndex = 0;
```



```
for (int i=0; i<4; i++) {
```

```
    if (s[i].marks < minMark) {
```

```
        minMark = s[i].marks;
```

```
        minIndex = i;
```

```
    }
```

```
}
```

```
float maxMarks = s[0].marks;
```

```
int maxIndex = 0;
```

```
for (int i=0; i<4; i++) {
```

```
    if (s[i].marks > maxMarks) {
```

```
        maxMarks = s[i].marks;
```

```
        maxIndex = i;
```

```
    }
```

```
}
```

14

```
printf ("Name \t ID \t Marks");  
for (int i=0; i<4; i++)  
    printf ("%s \t %d \t %f \n", s[i].name,  
        s[i].id, s[i].marks);  
}  
printf ("Min : %d", s[minIndex].marks);  
printf ("Max : %d", s[maxIndex].marks);  
}
```

4

Q

D:\\students directory में file create करके file में marks लिखें

50

30

10

90 → Sum

⑥

#include <stdio.h>

void main () {

float n1, n2, sum = 0;

FILE *fp;

fp = fopen ("input.txt", "r");

fscanf (fp, "%f %f", &n1, &n2);

sum = n1 + n2;

printf ("%f", sum);

fclose (fp);

}

16
5

d

a

Print → 5, 9

call change (&a, &b)

*x = 5 + 10 = 15

*y = 9 - 3 = 6

print → 15 6

<stdio.h> absolute the

f() output bioV
15 6
15 6

b

#include <stdio.h>

void calculate (int, int, int*, int*, int*, int*);

void main () {

int n1, n2, sum, diff, multi;

float quot;

scanf ("%d %d", &n1, &n2);

calculate (n1, n2, &sum, &diff, &multi, ");

printf ("%d %d %d %f", sum, diff, multi, quot);

}

void calculate (int n1, int n2, int *sum, int *diff, int *multi, int *quot) {

*sum = n1 + n2;

*diff = n1 - n2;

*multi = n1 * n2;

*quot = (float) n1 / n2;

}

Spring-2019

①

② step 1

num = 3459 > 0

rem = 9

sum = 0 + 9 = 9

num = 345

step 2

num = 345 > 0

rem = 5

sum = 9 + 5 = 14

num = 34

step 3

num = 34 > 0

rem = 4

sum = 14 + 4 = 18

num = 3

step 4

num = 3 > 0

rem = 3

sum = 18 + 3 = 21

num = 0

print →

rem	sum	num
3	21	0

num = sum = 21

18

steps

num = 21 > 0
rem = 1
sum = 21 + 1 = 22
num = 2

num = 2 > 0
rem = 2
sum = 22 + 2 = 24

num = 0
print → 24
num = 24

Infinite loop

⑥ #include <stdio.h>

#include <math.h>

void main()

int n, sum = 0, i = 0;

scanf("%d", &n);

while (n != 0)

int rem = n % 10;

sum += pow(2, i) * rem

i++;

n = n / 10;

printf("%d", sum);

②

①

step 1

main() में मिला

print → 10 20

step 2

func1(18.5) call

print → 18.500000

step 3

func2(23, 16.8) call

print → 23 16.800000

step 4

as a, b global variable

a = b = 10 + 20 = 30

print → 30 30

⑥

#include <stdio.h>

int add(int a, int b){

return a+b;

}

int sub(int a, int b){

return a-b;

}

int mult(int a, int b){

return a*b;

}

(20)

```
int div (int a, int b) {  
    if (b == 0) {  
        return -1;  
    }  
    return a/b;  
}
```

return a/b;

}

```
void main () {
```

```
    int n1, n2;
```

```
    char ch;
```

```
    scanf ("%d %d", &n1, &n2);
```

```
    scanf ("%c", &ch);
```

```
    switch (ch) {
```

```
        case '+':
```

```
            printf ("%d", add (n1, n2));
```

```
            break;
```


case '-':

printf("%d", sub(n1, n2));

break;



case '*':

printf("%d", mult(n1, n2));

break;

case '/':

printf("%d", div(n1, n2));

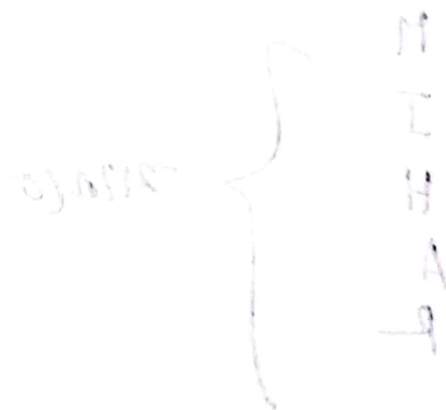
break;

default:

printf("Error");

}

}



22

3

Q

func(3)

print → 3

func(2)

print → 2

func(1)

print → 1

func(0)

return

1
2
3

3

2

1

1

2

3

or

error cause

fpr = fopen("dest.txt", "w");

then,

dest.txt file create 220 220 220

M

I

H

A

R

220/0

⑥

#include <stdio.h>

void main () {

FILE *fp; // Main, a - Main file

fp = fopen("emni.txt", "w");

int n;

for (int i=0; i<10; i++) {

scanf("%d", &n);

fprintf(fp, "%d", n);

}

}

④

#include <stdio.h>

typedef struct {

char name[50];

int id;

int mark;

char grade;

} student;

(24)

```
void main() {
```

```
    student s[100];
```

```
    int maxM = 0, minM = 1000, maxI, minI;
```

```
    for (int i = 0; i < 100; i++) {
```

```
        scanf("%s", s[i].name);
```

```
        scanf("%d", s[i].id);
```

```
        scanf("%d", s[i].mark);
```

```
        if (s[i].mark > 90) {
```

```
            s[i].grade = 'A';
```

```
        }
```

```
        else if (s[i].mark > 80) {
```

```
            s[i].mark = 'B';
```

```
        }
```

```
        else if (s[i].mark > 70) {
```

```
            s[i].mark = 'C';
```

```
        }
```



```
else {
```

```
    s[i].mark = 'F';
```

```
}
```

```
if (s[i].mark > maxM) {
```

```
    maxM = s[i].mark;
```

```
    maxI = i;
```

```
}
```

```
if (s[i].mark < minM) {
```

```
    minM = s[i].mark;
```

```
    minI = i;
```

```
}
```

```
}
```

```
printf("Max: %d", s[maxI].mark);
```

```
printf("\n Min: %d", s[minI].mark);
```

```
}
```

26

5

a

```
#include <stdio.h>
```

```
void main()
```

```
{  
    int *a, *b, *temp;
```

```
    a = (int*) malloc(sizeof(int));
```

```
    b = (int*) malloc(sizeof(int));
```

```
    scanf("%d %d", a, b);
```

```
    printf("%d %d\n", *a, *b);
```

```
    temp = a;
```

```
    a = b;
```

```
    b = temp;
```

```
    printf("%d %d\n", *a, *b);
```

```
    free(a);
```

```
    free(b);
```

```
}
```

⑥

print $\rightarrow 10$

change (8a)

 $*x = 10 * 11 = 110$ $++ *x = 111$ print $\rightarrow 111$

OR

print $\rightarrow 10 \quad 20 \quad 30 \quad 40 \quad 50$ func (A, 5)
 $\rightarrow$ pointerfunc 10 $\rightarrow$ 10

$$B[0] = 10 * (0+1) = 10$$

$$B[1] = 20 * (1+1) = 40$$

$$B[2] = 30 * (2+1) = 90$$

$$B[3] = 40 * (3+1) = 160$$

$$B[4] = 50 * (4+1) = 250$$

print $\rightarrow 10 \quad 40 \quad 90 \quad 160 \quad 250$

(28)

Summer - 2019

01 ← thing (d)

①
② strcpy (str1, "CSE") → str1 = "CSE"
strcpy (str2, "UIU") → str2 = "UIU"
i = strlen(str1) = 3

III = x" ++

III ← thing

step 1
i = 3, k = 0,
str1[3+0] = U

step 3
i = 3, k = 2
str1[5] = U

step 2
i = 3, k = 1
str1[4] = I

print → CSEUIU
print → new line
print → UIUES

⑥ #include <stdio.h>

void main()

int n;

scanf ("%d", &n);

int A[100][100];


```
for (int i=0; i<n; i++) {
```

```
    for (int j=0; j<n; j++) {
```

```
        scanf ("%d", &A[i][j]);
```

```
    }
```

```
for (int j=0; j<n; j++) {
```

```
    for (int i=0; i<n; i++) {
```

```
        if (A[i][j] % 2 == 0)
```

```
            printf ("%d", A[i][j]);
```

```
    }
```

```
    printf ("\n");
```

```
}
```

```
}
```

(30)

(2)

(a)

step 1print $\rightarrow 11, 25$ step 2

call func1(5.5)

print $\rightarrow 5.500000$ step 3

call func2(5, 4, 5)

print $\rightarrow 5$ step 4

call func2(12, 15.0)

print $\rightarrow 12 \quad 15.000000$ step 5

call func1()

$$\left. \begin{array}{l} a = 11 \\ b = 25 \end{array} \right\} b = 36$$
print $\rightarrow 36 \quad 36$

(b) #include <stdio.h>

int digitPos(int n) {

n = n / 10;

return n % 10;

}

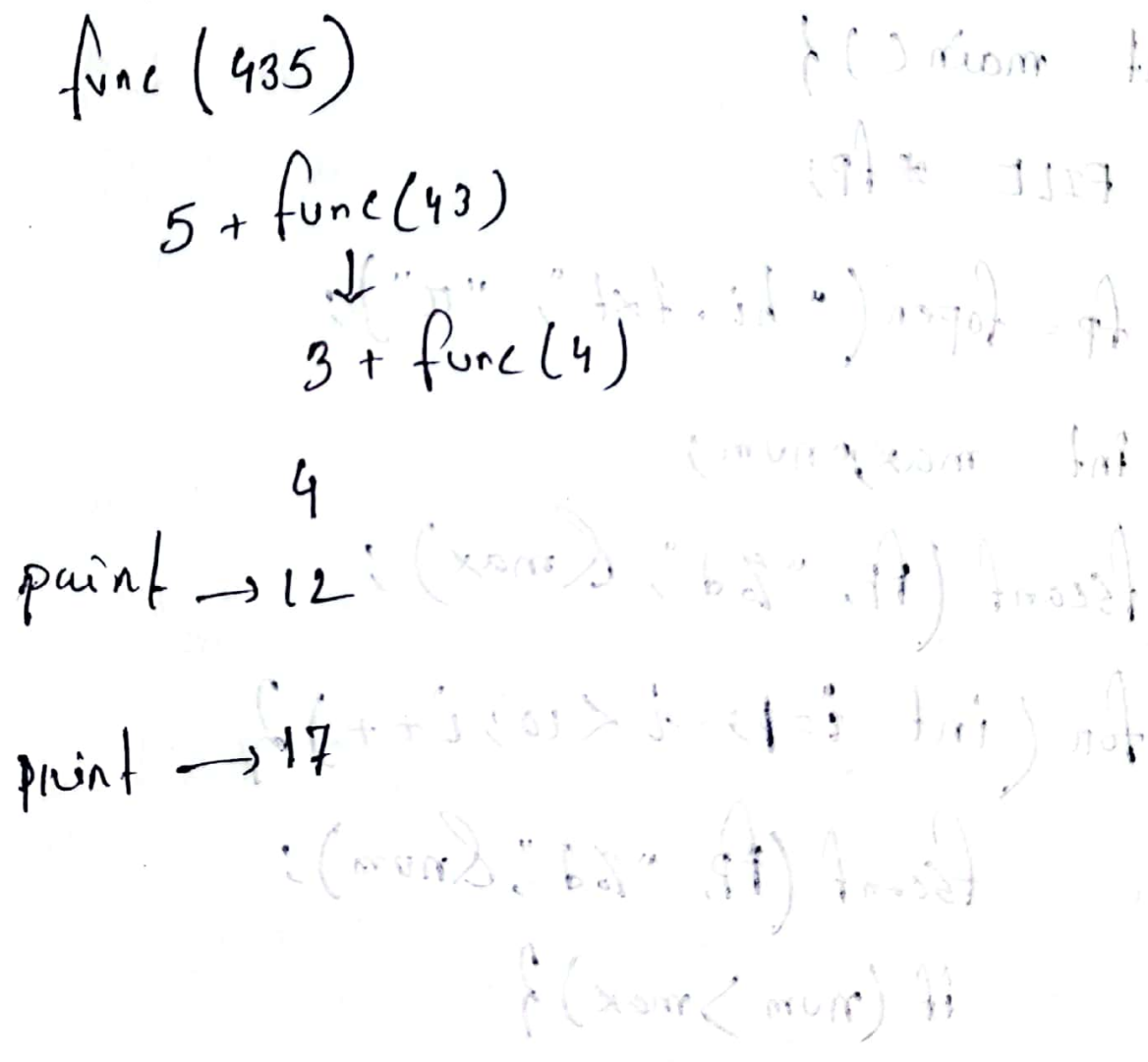
```

void main() {
    int num;
    scanf("%d", &num);

    int secondDig = digitPos(num);
    printf("%d", secondDig);
}

```

③
②



32

OR

D:\students\dest.14 file no. 14

50 50

40 90

30 120

20 140

10 150

⑥ #include <stdio.h>

int main()

FILE *fp;

fp = fopen("hi.txt", "r");

int max, num;

fscanf(fp, "%d", &max);

for (int i = 1; i < 10; i++)

fscanf(fp, "%d", &num);

if (num > max)

max = num;

}


```
fclose(fp);
printf("%d", max);
```

```
}
```

total = 0

④ #include <stdio.h>

```
typedef struct {
    char name[20];
    int id;
    float marks;
```

```
} students;
```

```
void main()
```

```
{
    students s[4];
```

```
    float avg, total = 0;
```

```
    for (int i = 0; i < 4; i++) {
```

```
        scanf("%s", s[i].name);
```

```
        scanf("%d", &s[i].id);
```

```
        scanf("%f", &s[i].marks);
```

(34)

```
total = total + s[i].marks; (q1) = total  
; (x000, "6.5") } printf  
}
```

avg = total / 4;

```
for (int i=0; i<4; i++) {  
    printf ("%s %d %f \n", s[i].name,  
        s[i].id, s[i].marks);  
}
```

```
printf ("Avg %f", avg);  
}
```

5
a) #include <stdio.h>

void main () {

int sum;

int *a = (int *) malloc (sizeof (int));

int *b = (int *) malloc (sizeof (int));

scanf ("%d %d", &a, &b); // (0 + 0) = [0] a

sum = *a + *b; // (1 + 1) = [1] a

printf ("%d", sum); // (2 + 0) = [2] a

free(a);

free(b);

}

6

print → ^a10, ^b20

change(&a, b)



*x = 10 + 11 = 21

print → 21, 20

or

print → 50 40 30 20 10

func(A, 5)

36

$$B[0] = (10 + 0) * (0 + 1) = 10$$

$$B[1] = (10 + 1) * (1 + 1) = 42$$

$$B[2] = (30 + 2) * (2 + 1) = 96$$

$$B[3] = (40 + 3) * (3 + 1) = 172$$

$$B[4] = (50 + 4) * (4 + 1) = 270$$

print → 10 42 96 172 270

Fall-2020

①
a

$$\text{int } a = 42 \% 5 = 2$$

$$\text{int } b = 42 / 5 = 8$$

print → ^b8 ^a2

step 1

call func1 (8 + 5 - 5)

print → 13.500000

step 2

call func2 (5, 4.5)

print → 5 4.5000000

step 3

call func2 (12, 15.0)

print → 12 15.000000

step 4

$$a = \text{func1}(8 + 5 \cdot 5)$$

$$\text{return } b = 8 * 2 = 16$$

return 15

print $\rightarrow$ 15 16

②

step 1

$$pp\ a = 42 \% 5 = 2$$

$$b = 42 / 5 = 8$$

$$c = 42$$

print $\rightarrow$ 2 8 42

③

$$\text{num}[3] = 42 \% 10 = 2$$

$$\text{num}[0] = 42 \% 100 = 42$$

$$\text{num}[4] = 42 \% 1000 = 42$$

$$\text{num}[1] = \text{num}[0] + \text{num}[3] = 42 + 2 = 44$$

$$\text{num}[2] = \text{num}[1] + \text{num}[4] = 44 + 42 = 86$$

$$0 == \&\&\&\& \&\& = [p] \text{ mem}$$

$$\&\& = \&\& + 0 = \text{mem}$$

[p] mem
 $\&\& \leftarrow$ bring [p] or [p] + 1step 3

$$0 == \&\&\&\& \&\& = [s] \text{ mem}$$

step 2

call change(&a, &b, c)

$$*x = 2 + 10 = 12$$

$$*y = 8 + 3 = 11$$

print $\rightarrow$ 12 11 42

$$\&\&\& = \&\& + p \&\& = \text{mem}$$

 $\&\& \leftarrow$ bring [p] or [p] + 1step 4

sp

s

p

pp

sp

sis

38

step 1

$$\text{num}[4] = 42 \% 2 == 0$$

$$\text{sum} = 0 + 42 = 42$$

file se value print $\rightarrow$ ^{num[4]} 42

step 3

$$\text{num}[2] = 86 \% 2 == 0$$

$$\text{sum} = 44 + 86 = 130$$

file se value print $\rightarrow$ 86

step 5

$$\text{num}[0] = 42 \% 2 == 0$$

$$\text{sum} = 174 + 42 = 216$$

file se value print $\rightarrow$ 42

output

42

2

86

44

42

216

step 2

$$\text{num}[3] = 2 \% 2 == 0$$

$$\text{sum} = 42 + 2 = 44$$

file se value $\rightarrow$ ~~num[3]~~ 2

step 4

$$\text{num}[1] = 44 \% 2 == 0$$

$$\text{sum} = 130 + 44 = 174$$

file se value print $\rightarrow$ 44

step 6

file se value print $\rightarrow$ 216

④

```
#include <stdio.h>
```

```
int digitMult (int id) {
```

```
int ans  
sum = 1;
```

```
int ans = 1;
```

```
while (id != 0) {
```

```
    int digit = id % 10;
```

```
    ans = ans * digit;
```

```
    id = id / 10;
```

```
}
```

```
return ans;
```

```
}
```

```
void main ( ) {
```

```
    int id = 42;
```

```
    int product = digitMult (id);
```

```
    printf ("%d", product);
```

```
}
```

```

5) #include <stdio.h>
    #include <string.h>
    void main

```

```

    <stdio.h>
    <string.h>

```

```

    typedef struct

```

```

    {() name bio

```

```

        char name[50];

```

```

        int id;

```

```

        char phone[20];

```

```

        char birthdate[20];

```

```

    } person;

```

```

    void main() {

```

```

        person myself = { "Slakib", 11202018, "012-...",

```

```

        "20-05-2003" }

```

```

        int birthyear = atoi(strchr(myself.birthdate, '-') + 1);

```

```

        printf("My name is: %s\n", myself.name);

```

```

        printf("    id : %d\n", myself.id);

```

```

        printf("My Birthyear: %d", birthyear);

```

```

    }

```


4.1

⑥

```
#include <stdio.h>
#include <string.h>
```

```
void main()
```

```
char name[20] = "Roni";
```

```
char id[20] = "011202018";
```

```
char ans[50];
```

```
strcpy(ans, name);
```

```
strcat(ans, " ");
```

```
strcat(ans, id);
```

```
puts(ans);
```

Summer-2021

42

①
int a, b;
float e;

step 1

$a = 2042 \% 39 = 14$

$b = \text{func1}(14)$

$b = 56.000$

$\text{func2}(a, b)$

step 2

call $\text{func1}(14)$

$c = 14 + 14 = 28$

$\text{func2}(c, b)$

step 5

call $\text{func2}(14, 56.00)$

$*x = 14 * 2 = 28$

$y = 28.00$

step 3

call $\text{func2}(28, 0.00)$

$*x = 28 * 2 = 56$

$y = 14.000$

④
②

$$\text{int } a = 2042 \% 5 = 2$$

$$b = 2042 \% 4 = 2$$

$$c = 2042 \% 6 = 2$$

strcpy(my_str, arr[2]) → my_str = Steve Wozniak

my_str[2] = toupper(my_str[2]) → E

strcpy(temp_str, arr[2], 2) → temp_str = St

strcat(my_str, temp_str);

puts(my_str) → StEve WozniakSt

③

Hello Vaxxens

2

2

2

2

2042

→ a = 2042

$$a \% 10 = 2$$

$$a \% 20 = 2$$

$$a \% 30 = 2$$

$$a \% 40 = 2$$

④

$$\text{int } b = 2042 \% 11 = 7$$

$$a[5] = \{8, 9, 10, 11, 12\}$$

$$t = 8$$

$$u = 9$$

$$v = 9 \rightarrow *++p1 \rightarrow \text{2nd memory address}$$

$$w = 10 \rightarrow *(++p2) \rightarrow \text{3rd}$$

⑥

44

```
#include <stdio.h>
```

```
int main()
```

```
int idValues[4] = {0};
```

```
void takeInput();
```

```
int elementProd (int *, int);
```

```
void main () {
```

```
    takeinput();
```

```
    int *ptr = idValues;
```

```
    int ans = elementProd (ptr, 4);
```

```
    printf ("%d", ans);
```

```
}
```

```
void takeInput () {
```

```
    for (int i=0; i<4; i++) {
```

```
        scanf ("%d", &idValues[i]);
```

```
    }
```

```
}
```


45

```
int elementProd (int *arr, int size) {
```

```
    int prod = 1;
```

```
    for (int i = 0; i < size; i++) {
```

```
        prod *= arr[i];
```

```
    }
```

```
    return prod;
```

```
}
```

```
}
```

6

```
#include <stdio.h>
```

```
#include <string.h>
```

```
void main () {
```

```
    char str_a [2042 % 11 + 30];
```

```
    char str_b [strlen(str_a)];
```

```
    int i = 0, flag = 0;
```

```
    gets (str_a);
```

```
for (int i=0; str_a[i] != '\0'; i++) {  
    if (str_a[i] >= '0' && str_a[i] <= '9') {  
        flag = 1;  
        str_b[i++] = str_a[i];  
    }  
}
```

```
if (flag == 0) {  
    printf("0");  
}
```

```
else {  
    puts(str_b);  
}
```

3.

47

7

```
#include <stdio.h>
```

```
typedef struct {
```

```
    char student_ID[10];
```

```
    char student_Name[50];
```

```
    float marks[5];
```

```
} student_info;
```

```
void main () {
```

```
    student_info students[2042%11+10];
```

```
    for (int i=0; i < size; i++) {
```

```
        strcpy (students[i].student_Name, "Ihinku");
```

```
        strcpy (students[i].student_ID, "042");
```

```
        for (int j=0; j < 5; j++) {
```

```
            students[i].marks[j] = 0;
```

```
        }
```

```
}
```

```
for (int i = 0; i < size; i++) {
```

```
    fflush(stdin);
```

```
    gets (students[i].student_ID);
```

```
    fflush(stdin);
```

```
    gets (students[i].student_Name);
```

```
for (int j = 0; j < 5; j++) {
```

```
    scanf ("%f", &students[i].marks[j]);
```

```
    }
```

```
}
```

```
}
```


Fall - 2021

49

①

step 1

$$a = 2042 \% 47 = 21$$

$$b = 0$$

$$c = 0$$

call func3(a, &b)

~~b = 1~~

$$b = 1 * 1 = 1$$

~~step 2~~

$$a = 21$$

$$b = 1$$

$$c = 0$$

step 2

$$b = \text{func1}(a)$$

call func1(a)

$$c = 21 + 21 = 42$$

call func2(&a, b)

$$*x = 21 * 2 = 42$$

$$b = \text{func3}(a, x)$$

return 43

step 3

func2(&a, b)

$$*x = 43 * 2 = 86$$

$$b = \text{func3}(a, x)$$

return 87

$$a = 87 < U$$

$$U = b = 42$$

$$c = 42$$

memory address ++ 20 so *n
a = 43
b = 42
c = 42

50

2

1505-1107

1

a = 2042 % 5 = 2

b = 2042 % 4 = 2

c = 2042 % 3 = 2

stnepy (stn1, arr[2]) -> stn1 = UIU is the Best
stnepy (stn2, arr[4%5]) -> stn2 = Be a problem solver

U > B

stn1[2] = U

stnepy (stn2, arr[2], 2) -> stn2 = UI a problem solver.

stnecat (stn1, stn2) -> UIU is the Best UI a problem solver

3

Hello 3 doses!!!

Line 4: 2050, 2046, 2042,

Line 5: 2049, 2045,

Line 6: 2048, 2044,

Line 7: 2051, 2047, 2043

4

```
char a[] = "Partho Majumdar"
n
r
d ar
um d ar
ajum d ar
Majum d ar
h o Majum d ar
rtho Majum d ar
Partho Majum d ar

{ (0 == 15) }
char *b = a
for (int i=0; i<n; i++, b++)
    b = (15-i)
b -= 2 -> 15
13
11
9
}
```

5

```
#include <stdio.h>

void main() {
    char a[] = "Partho Majumdar";
    char *p = a;
    revNam(p, 15);
}
```

52

```
void revName (int *p, int len) {  
    if (len == 0) {  
        return;  
    }  
    else {  
        printf ("%c", p[len-1]);  
        revName (p, len-1);  
    }  
}
```

⑥

```
#include <stdio.h>
```

```
void main () {
```

```
    int scores [6][45];
```

```
    int initVal = 2042 % 11 + 10;
```



```
for (int i=0; i < 6; i++) {
```

```
    for (int j=0; j < 45; j++) {
```

```
        scores[i][j] = initVal;
```

```
    }
```

```
}
```

```
for (int i=0; i < 45; i++) {
```

```
    scores[2][i] += i * 10;
```

```
}
```

↘ 'c' section

```
for (int i=0; i < 6; i++) {
```

```
    printf("sec: %c", 'A'+i);
```

```
    for (int j=0; j < 45; j++) {
```

```
        printf("%1 ", scores[i][j]);
```

```
    }
```

```
    printf("\n");
```

```
} }
```

⑧

#include <stdio.h>

#include <string.h>

typedef struct {

char name[100];

char email[100];

char password[100];

int testMark;

} Candidate;

void login (Candidate candidates[], int size) {

char email2[100];

char password2[100];

scanf ("%s %s", &email2, &password2);

for (int i = 0; i < size; i++) {

if (strcmp (candidate[i].email, email2) == 0

&& strcmp (candidate[i].password, password2) == 0)
 {

55

```
printf ("%d " Candidates [i]. test_marks);  
return;
```

```
}
```

```
}
```

```
printf  
}
```

```
Void main ( ) {
```

```
int size = 2042;
```

```
candidate candidates [2042-size];
```

```
for (int i=0; i < size; i++) {
```

```
scanf ("%s", candidates[i].name);
```

```
scanf ("%s", candidates[i].email);
```

```
scanf ("%s", candidates[i].password);
```

```
scanf ("%d", & candidates[i].testMarks);
```

```
}
```

```
login (candidates, size);
```

```
}
```

8

56

```
#include <stdio.h>
```

```
int search (int arr[], float value, int n, int r) {
```

```
    int count = 0;
```

```
    for (int i = 0; i < n; i++) {
```

```
        if (arr[i] == value) {
```

```
            count ++;
```

```
        if (count == n) {
```

```
            return i;
```

```
        }
```

```
    }
```

```
}
```

```
return -1;
```

```
}
```


(57)

void main() {

int n, r, value; ^{occurrence}
scanf("%d %d %f", &n, &value);

float arr[n];

for (int i = 0; i < n; i++)
scanf("%f", &arr[i]);

}

int index = search(arr, value, n, r);

if (index != -1) {

printf("%d", index);

}

else {

printf("Not found");

}

}

Spring-21

58

①

step 1 $a = r + c = \text{num}$

$$\text{num1} = 2042 \% 7 = 5$$

$$\text{num2} = f1(\text{num1})$$

step 2

$$\text{num2} = 5 * 5 = 25$$

$$\text{return } 25 - 1 = 24$$

②

step

$$a = 2042 \% 8 = 2 ; \text{str1} = "011222042"$$

$$\text{strcat}(\text{str1}, " "); \rightarrow 011222042$$

$$\text{strcat}(\text{str1}, \text{arr}[2]); \rightarrow 011222042 \text{ is friendly}$$

$$\text{strcpy}(\text{str1}, \text{strcat}("s")); \rightarrow s \text{ friendly}$$

57

3

$$a = 2042 \% 7 = 5$$

step 1

$$i = 0, i < 5$$

$$\text{num}[0] = (2 \times 0) + 5 = 5$$

step 2

$$i = 1, i < 5$$

$$\text{num}[1] = (2 \times 1) + 5 = 7$$

step 3

$$i = 2, i < 5$$

$$\text{num}[2] = (2 \times 2) + 5 = 9$$

steps

$$i = 3, i < 5$$

$$\text{num}[3] = (2 \times 3) + 5 = 11$$

step 6

$$i = 4, i < 5$$

$$\text{num}[4] = (2 \times 4) + 5 = 13$$

step 7

$$i = 0, i < 4$$

$$\text{sum} = 0 + 5 = 5$$

step 8

$$i = 1, i < 4$$

$$\text{sum} = 5 + 7 = 12$$

step 9

$$i = 2, i < 4$$

$$\text{sum} = 12 + 9 = 21$$

step 10

$$i = 3, i < 4$$

$$\text{sum} = 21 + 11 = 32$$

txt

Good Morning

5

7

9

11

32

④

5
14
23
32
18

⑤

#include <stdio.h>

int sumofSeven (int *arr, int n) {

int sum = 0;

for (int i = 0; i < n; i++)

if (arr[i] % 7 == 0)

sum += arr[i];

}
}

} return sum;

<stdio.h>
$\sqrt{2}$

f() return

61

```
int main () {
```

```
    int scores[5]
```

```
    for (int i=0; i<5; i++) {
```

```
        scores[i] = (2042 % 9) + 2 * i;
```

```
    }
```

```
    int *ptr = scores;
```

```
    int sum = sumofSeven(ptr, 5);
```

```
    printf("%d", sum);
```

```
}
```

62

```
#include <stdio.h>
```

```
#include <string.h>
```

```
void main () {
```

```
    char sentence[100];
```

```
    char id[] = "011222042";
```

```
    gets(sentence);
```

```
for (int i=0; sentence[i] != '\0'; i++) {  
    if (i==0 || sentence[i-1] == ' '){  
        sentence[i] = toupper(sentence[i]);  
    }  
    else {  
        sentence[i] = tolower(sentence[i]);  
    }  
}
```

```
strcat(sentence, ".");
```

```
strcat(sentence, " ");
```

```
puts(sentence);
```

```
}
```

(63)

⑦

```
#include <stdio.h>

typedef struct {
    char id[20];
    int age;
    float weight;
} Soldier;
```

```
Soldier takeInput (Soldier sol) {
    scanf("%s %d %f", sol.id,
          sol.age, (sol.weight));
    return sol;
}
```

```
void main() {
    Soldier sol1;
    sol1 = takeInput(sol1);
    printf("%s %d %f", sol1.id, sol1.age, sol1.weight);
}
```

8

69

```
#include <stdio.h>
```

```
#include <string.h>
```

```
void main()
```

```
{  
    char sentence[100];
```

```
    int count = 0;
```

```
    gets(sentence);
```

```
    for (int i = 0; sentence[i] != '\0'; i++)
```

```
    {  
        if (isupper(sentence[i]))
```

```
        {  
            count++;
```

```
        }  
    }
```

```
}
```

```
if (count % 2 == 0)
```

```
{  
    printf("011222042");
```

```
}
```

```
else
```

```
{  
    printf("Jhinku");
```

```
}
```

```
}
```

```
#include <stdio.h>
```

```
void main()
```

```
{  
    char str[100];  
    gets(str);
```

```
    int count = 0;
```

```
    for (char *ptr = str; *ptr != '\0'; ptr++)
```

```
    {  
        if (*ptr >= 'A' && *ptr <= 'Z')  
        {  
            count++;
```

```
        }  
    }
```

```
    if (count % 2 == 0)
```

```
    {  
        printf("011222666");
```

```
    }  
    else  
    {  
        printf("Jhinku");
```

```
    }
```